

7.2 CRITICAL VALUES

Table 7.1 lists common critical values for a standard normal random variable, z_α , defined by (see Figure 7.2):

$$\text{Prob}[Z \geq z_\alpha] = \alpha. \quad (7.3)$$

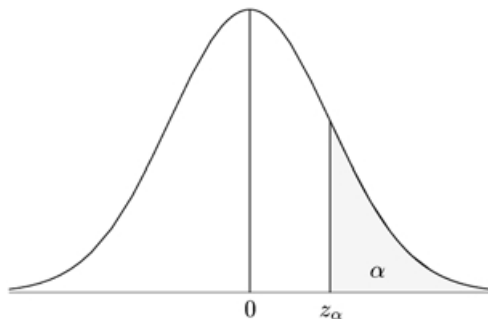


Figure 7.2: Critical values for a normal random variable.

α	z_α	α	z_α	α	z_α
.10	1.2816	.00009	3.7455	.000001	4.75
.05	1.6449	.00008	3.7750	.0000001	5.20
.025	1.9600	.00007	3.8082	.00000001	5.61
.01	2.3263	.00006	3.8461	.000000001	6.00
.005	2.5758	.00005	3.8906	.0000000001	6.36
.0025	2.8070	.00004	3.9444		
.001	3.0902	.00003	4.0128		
.0005	3.2905	.00002	4.1075		
.0001	3.7190	.00001	4.2649		

Table 7.1: Common critical values

7.3 TOLERANCE FACTORS FOR NORMAL DISTRIBUTIONS

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n from a normal population with mean μ and standard deviation σ . Using the summary statistics $\bar{x}$ and s , a tolerance interval $[L, U]$ may be constructed to capture 100P% of the population with probability $1 - \alpha$. The following procedures may be used.

- (1) Two-sided tolerance interval: A $100(1 - \alpha)\%$ tolerance interval that captures 100P% of the population has as endpoints

$$[L, U] = \bar{x} \pm K_{\alpha, n, P} \cdot s \quad (7.4)$$